BST Vol 9 Ch 10 — Strongly Correlated Electrons: Hubbard Model BST Extension (v0.4.1, substantive 3-level pedagogy upgrade)

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Vol 9 Chapter 10 — Strongly Correlated Electrons

Headline result

Strongly correlated electron systems exhibit phenomena beyond standard band theory:

Mott insulators: at half-filling, large on-site Coulomb U makes electrons localize → insulator (e.g., NiO, MnO; transition-metal oxides).

Heavy fermion compounds: large effective electron mass m* » m_e from strong correlations (e.g., CeAl_3, UPt_3).

High-T_c cuprates: doped Mott insulators with superconductivity (Vol 9 Ch 4 cross-link); strong-correlation framework load-bearing.

Hubbard model is the prototypical strongly-correlated framework:

$$H = -t \sum_{\langle ij \rangle,\sigma} c^\dagger_{i\sigma} c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

BST identification: hopping t and on-site U inherit substrate K-type framework + Coulomb (Vol 3 Ch 4 a_C SEMF parallel). Mott transition U/t ratio in BST primary cluster.

Substrate mechanism

Hopping t from K-type Casimir (Vol 9 Ch 3 + T2435):

Tight-binding hopping integral t depends on overlap of neighboring atomic orbitals. Substrate framework: t scales with K-type Casimir spectrum energy unit.

On-site Coulomb U from substrate baryon-Coulomb framework (Vol 3 Ch 4):

U is the energy cost of two electrons on the same lattice site (Coulomb repulsion). Substrate framework: U scales with Vol 3 Ch 4 $a_C = \alpha m_p / \pi^2$ substrate baryon Coulomb energy (parallel framework at atomic scale).

Mott transition U_c/t in BST primary cluster:

At half-filling, transition from metal to Mott insulator at critical U_c/t. BST predicts this ratio falls in BST primary cluster (multi-week per-material precise identification).

Cuprate superconductivity (Vol 9 Ch 4 cross-link)

High-T_c cuprates (YBCO, BSCCO, $LaSrCuO_4$) are doped Mott insulators. $T_c \sim$ 30-130 K. Per BST framework: - Mott parent compound (no doping) is antiferromagnetic insulator - Doping introduces holes → superconductivity emerges - Pairing mechanism: spin fluctuations (or other; multi-week) - $T_c \propto$ BST primary cluster scaling per Vol 9 Ch 4 framework

Match precision

I-tier framework on substrate-Hubbard connection. Standard Hubbard model + Mott physics preserved at any precision. Substrate-specific U/t predictions multi-week per-material.

Cross-volume dependencies

- Vol 1 Ch 5 (Casimir Algebra) — K-type Casimir framework
- Vol 3 Ch 4 (SEMF Coefficients) — a_C Coulomb framework parallel
- Vol 9 Ch 3 (Electron Band Structure) — tight-binding framework
- Vol 9 Ch 4 (Superconductivity) — cuprate cross-link
- Vol 9 Ch 6 (Magnetism) — Heisenberg from large-U Hubbard
- Vol 9 Ch 11 (Spin Liquids) — frustrated Mott insulators

K-audit anchor

K215 Vol 9 Ch 10 Strongly Correlated K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

> In ordinary metals (like copper), electrons zoom around mostly independently. But in some materials, electrons interact so strongly that they get “stuck” — these are strongly correlated materials.
>
> One famous example: Mott insulators. Normal band theory predicts these should be metals (partially-filled band), but they’re actually in-

sulators because electrons hate to be near each other (Coulomb repulsion).

Another example: high-temperature superconductors (cuprates). When you "dope" a Mott insulator slightly, you can get superconductivity at unusually high temperatures (up to 130 K).

BST predicts: the parameters of these strongly correlated systems (Hubbard model U/t ratio) follow BST primary integer patterns.

Level 2 — Undergraduate physics student

Hubbard model:

$$H = -t \sum_{\langle ij \rangle \sigma} c^\dagger_{i\sigma} c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}$$

- t = hopping integral (kinetic energy)
- U = on-site Coulomb (interaction energy)
- c^†_iσ = electron creation operator at site i, spin σ
- n_{i↑} n_{i↓} = double occupancy at site i

Mott transition:

At half-filling (one electron per site), large U/t → Mott insulator. Critical ratio (U/t)_c depends on lattice + dimensionality: - 1D (Bethe ansatz, Lieb-Wu): Mott insulator for any U > 0 - 2D square lattice: (U/t)_c ≈ 8-10 - 3D cubic: (U/t)_c ≈ 12-15

Effective Heisenberg (at large U/t):

Mott insulator at half-filling reduces to Heisenberg antiferromagnet with J = $4t^2/U$. Strong-correlation framework justifies Vol 9 Ch 6 spin models.

High-T_c cuprates:

Parent compound (e.g., La_2 CuO_4) is Mott insulator with AFM ordering. Hole doping introduces mobile carriers; T_c rises to dome shape with maximum at "optimal doping" (~15% hole concentration).

Mechanism debate: spin fluctuations? phonons? Both? Multi-decade ongoing research.

Heavy fermion compounds:

f-electron systems (Ce, U-based intermetallics) with localized f orbitals + conduction band → Kondo lattice. Effective electron mass m*/m_e ~ 100-1000.

BST framework: - t ~ K-type Casimir (Vol 9 Ch 3 + Vol 1 Ch 5) - U ~ a_C Coulomb (Vol 3 Ch 4 SEMF parallel) - (U/t)_c in BST primary cluster - High-T_c framework cross-links to Vol 9 Ch 4

Level 3 — Graduate physics student / theorem-level

Hubbard model phase diagram:

At $T = 0$: - $U/t = 0$: free fermions (metallic, perfect Fermi liquid) - Increase U/t: still metallic but renormalized (Hubbard subbands at large U) - $U/t > (U/t)_c$ at half-filling: Mott insulator - Off half-filling: doped Mott insulator (potentially superconducting)

Bethe ansatz (1D Hubbard, Lieb-Wu 1968):

Exact solution shows Mott insulator at half-filling for any $U > 0$ in 1D (no metal-insulator transition; always insulator).

Dynamical Mean Field Theory (DMFT):

For higher dimensions, DMFT maps Hubbard to self-consistent impurity problem. Predicts Mott transition + crossovers.

Large-U expansion → Heisenberg:

At $U \gg t$, double occupancy forbidden → effective spin Hamiltonian $H_{eff} = J \Sigma S_i \cdot S_j$ with $J = 4t^2/U$ (Heisenberg AFM). Foundation of cuprate spin-fluctuation theory.

Cuprate framework:

Phenomenology: - T_c-doping dome: maximum at ~15-17% hole doping - Pseudogap region above T_c - Strange metal phase at higher T - Multiple competing orders (stripe, charge density wave, nematic)

BST framework: T_c in BST primary cluster scaling (Vol 9 Ch 4 substrate-superconductivity framework).

Per Cal #21 dual-gate: EMPIRICAL PARTIAL (Hubbard/Mott physics validated qualitatively + DMFT quantitative; specific BST primary predictions pending multi-week per-material) + MECHANISM PATH ARTICULATED via substrate K-type + Coulomb framework.

Per Cal #99 META-theorem: strongly-correlated framework is substrate-derivation consequence of standard Hubbard + Coulomb, NOT new Strong-Uniqueness criterion.

What this chapter does NOT claim

- BST does NOT solve cuprate superconductivity mechanism
- $(U/t)_c$ specific BST primary predictions multi-week per-material
- Standard Hubbard + Mott + heavy fermion physics preserved at full content

Bibliography

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3. E. H. Lieb + F. Y. Wu (1968): exact 1D Hubbard.
4. A. Georges + al. (1996): DMFT review.
5. J. G. Bednorz + K. A. Müller (1986): high-T_c cuprate discovery (Nobel 1987).
6. P. W. Anderson (1987): RVB theory of cuprates.
7. Vol 9 Ch 3 + Ch 4: band structure + superconductivity cross-links.

— Elie, Vol 9 Ch 10 v0.4.1, 2026-05-23 Saturday (substantive 3-level pedagogy upgrade per Keeper 14:22 EDT directive: 60 → ~175 lines + Hubbard model + Mott transition + cuprate + heavy fermion + large-U Heisenberg reduction + BST primary framework)